

(ALEX) TUNG NHI TRAN

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SUMMARY

AI Developer with 4+ years designing and deploying production LLM, agentic AI, and machine learning systems end-to-end. Proven track record taking AI solutions from concept to production, including RAG pipelines, intelligent automation, and multi-agent systems, with a strong commitment to responsible AI deployment, governance, and privacy by design. Experienced translating complex business requirements into practical AI solutions across technical and non-technical stakeholders. Master of IT (AI), UNSW. IBM RAG & Agentic AI certified.

SKILLS

AI & Machine Learning	PyTorch, TensorFlow, OpenCV, Computer Vision, NLP, RAG.
LLM & GenAI	RAG, MCP, prompt engineering, chatbot development.
Full-Stack Development	Flask, FastAPI, Gradio, HTML, CSS, JavaScript, Python, C++, R.
Cloud & DevOps	Docker, Ubuntu, Git/GitHub CI/CD workflows.
Data & Databases	SQL, ChromaDB, Vector Databases, Seaborn, Matplotlib, Numpy, Scipy.

EXPERIENCE

AI Engineer HeartBug Pty Ltd	April 2026 - Now <i>Sydney, Australia</i>
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- **Designed and implemented** end-to-end AI architecture for a production healthcare call centre — evaluating and selecting technology options across LLM providers, orchestration frameworks (MCP), and cloud infrastructure (AWS) to deliver a scalable, governed agentic AI solution with hybrid intent routing, identity verification, and slot filling
- **Designed end-to-end agentic AI pipeline** covering intent classification, context management, and safe response generation for high-stakes healthcare device queries.

Research Assistant Université Saint-Louis - Bruxelles	Aug 2021 - Sep 2022 <i>Brussels, Belgium</i>
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- **Conducted exploratory data analysis** (EDA) using R-based visualisation tools to identify patterns, trends, and variable relationships between synthetic BEAMM and real Belgian tax datasets (~14,000 households), generating insights to inform economic modelling decisions, with data privacy requirements when handling sensitive data.
- **Collaborated with economists** to identify demands and implement comments on variable interdependencies, leading to improved synthetic datasets for economic modeling.

AI Researcher & Lecturer Dong A University	Oct 2020 - Sep 2022 <i>Da Nang, Vietnam</i>
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- **Built and operationalized** 2 production ML systems for anomaly identification, VAE-based predictive maintenance and AlexNet + Isolation Forest for real-time visual anomaly detection. Achieving 97%+ accuracy, with continuous production monitoring and incident response to maintain system reliability.
- **Collaborated with cross-functional stakeholders** to define ML requirements, balance competing priorities, and deliver compliant solutions. Translating business and policy constraints into technical specifications for end-to-end ML pipelines covering data processing, model training, evaluation, and production monitoring.
- **Lectured** Introduction to Programming and supporting Calculus/Linear Algebra courses

Python Developer TMA Solutions	Jan 2020 - Mar 2020 <i>HCM, Vietnam</i>
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- **Developed scalable Python applications** with focus on code maintainability, reliability, and performance optimization for production environments

PROJECTS

Anomaly Detection in Smart Manufacturing | Published Research

- **Designed and deployed** end-to-end production computer vision pipeline for anomaly detection, achieving 97% accuracy.
- **Published findings** in "Machine Learning and Probabilistic Graphical Models for Decision Support Systems," CRC Press, 2022, pp. 34-61

Medical Image Segmentation - Breast Cancer Detection

- **Implemented U-Net architecture** for breast cancer detection in ultrasound images as part of UNSW Computer Vision coursework.
- **Optimized U-Net architecture** through channel hyperparameter tuning where varying channel numbers directly affected encoder-decoder performance for tumor detection, achieving 95.80% accuracy.

LLM Q&A Chatbot API - Flask/Gradio

- **Built production RAG architecture** using ChromaDB vector database, Multilingual E5 Large embeddings, and IBM Granite LLM via API, achieving <5s latency for context-aware Q&A.
- **Implemented FastAPI + Gradio interface** for end-to-end query processing, applying prompt filtering and output validation techniques to ensure safe and reliable responses.

Real-Time Emotion Detection System | University Thesis, 2019

- **Developed a real-time CNN-based emotion recognition system** in Python and C++ using OpenCV, integrating classical ML (Haar-like features, SVM) with deep learning inference in a production-quality pipeline.
- **Presented research** at 9th Scientific Research Conference of PFIEV - HCMUT Students, 2019.

EDUCATION

Master of IT, Major in AI, University of New South Wales (UNSW) Jun 2023 - Jun 2025
Relevant Coursework: Computer Vision, Natural Language Processing (Distinction).

B.E. (Telecommunications) of PFIEV, Ho Chi Minh University of Technology (HCMUT) Sep 2014 - Sep 2019
Training Program of Excellent Engineers in Vietnam, cooperating with France - PFIEV.
Focus Areas: Digital Signal Processing, Embedded Signal Systems.

ACHIEVEMENTS & PUBLICATIONS

Published Research: [Machine Learning and Probabilistic Graphical Models for Decision Support Systems \(CRC Press, 2022\)](#), [BME7 - Springer IFMBE Proceedings](#).

International Student Award: 15% tuition scholarship at UNSW.

International Engineering Program: Degree Appendix co-signed by HCMUT and IMT Atlantique (France)

Certifications: IBM RAG and Agentic AI Specialization, Vision Language Models (VLM) Bootcamp - OpenCV University (100%, 2025)